AIML-27-10-05-26

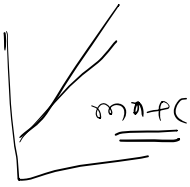
10 May 2026 09:00

① $\boxed{ML}$ Avatars ② ML and practical issues

$y = f(x)$

0	1	2	3	4
1	3	5	7	?

$2x+1$

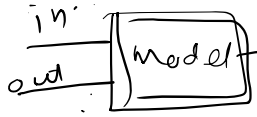


$mx+c$

A branch of AI:

Design and development of algos:

Capture and model behaviors:



$\rightarrow f(x)$

Based on emperical data:

Knowledge:

①

② → Hard coded rules (the old way) –

→ learning from data (the ML way)

③ Data is Knowledge.

Why now

2010–2015

① Data explosion: The internet, Sensors, IoT,

2005 : ~150 exabytes

2025 : ~180+ zettabytes (1 ZB = 1,000 EB)

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② Compute explosion:

③ Algorithms matured:

① ML → Arthur → 1959

Explicit programming	ML learning
if x > 5 : return "big"	Show 1000 examples of big/small → system infers the threshold
you write the logic	logic emerges from data
Doesn't improve with more data	Improves with more data

Kevin murphy → (2012)

① Training →
② Inferring →

Tom michel →

Algorithms that improve their performance (P) at some task (T) with experience (E)

T, P, E

what's the task (T)

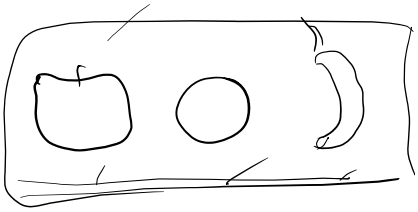
what's the task (T)

How will you measure performance (P)

What is experience data (E)

Spam filter: classify email spam or ham

- percentage of emails classified correctly
- database of emails



feature extraction

Shape	Size	color	variety
curved	small	yellow	banana

name gender address experience Qualification salary

experience	Qualification	Salary

learned

raw Input → [Feature extraction] → 2 features → [Algos] learned model

experience → [lm] → Salary

sentiment analysis.

→ imdb

→ rotten tomatoes

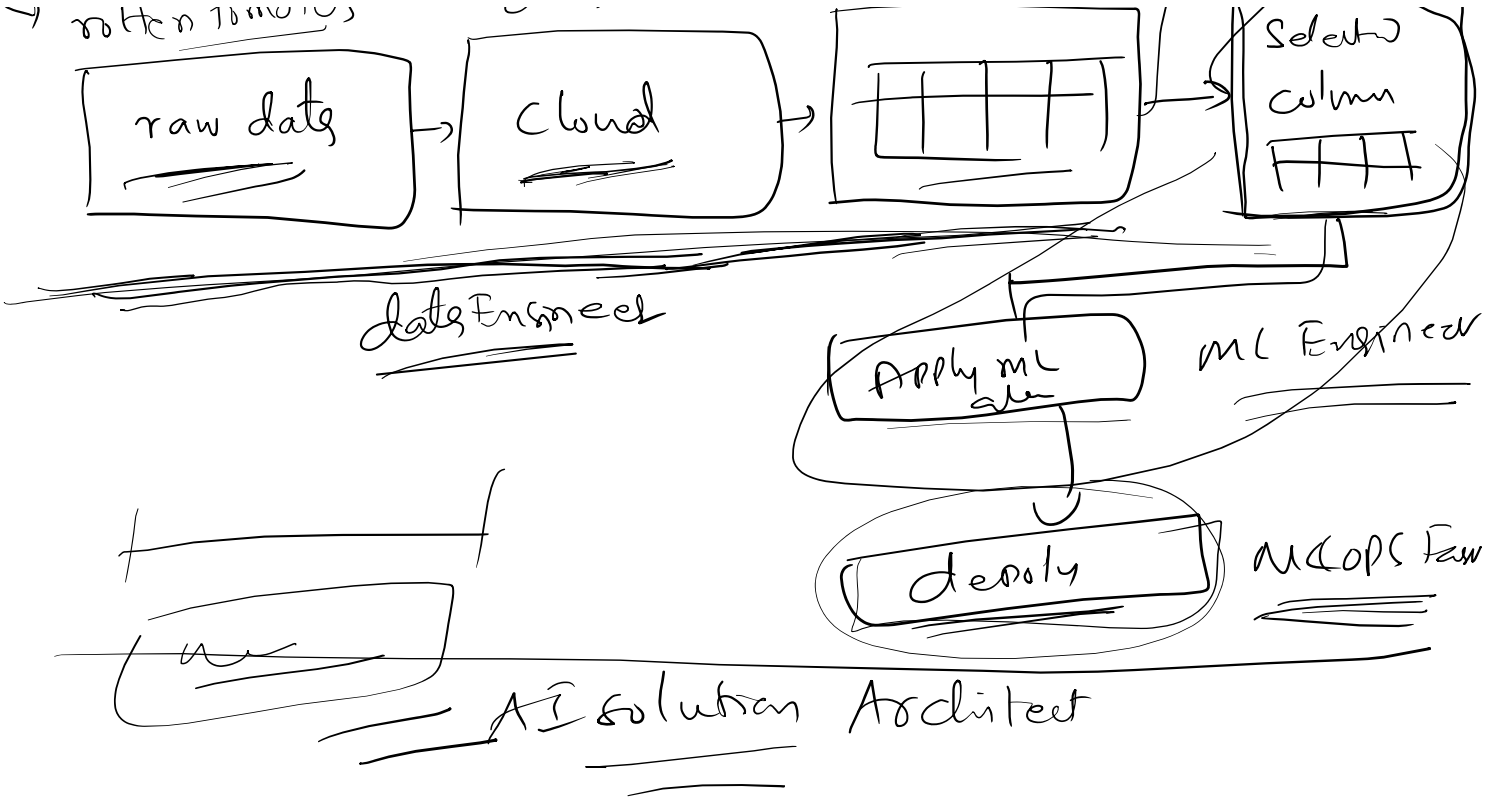
data lake

data preprocessor

data Analysis

Cleaning

Selection



Characteristics of good ML problems:

① Clear use case:

vague: Improve customer experience.

Concrete:

② Relevant data: ML needs labeled data

③ Decision making:

2 way split

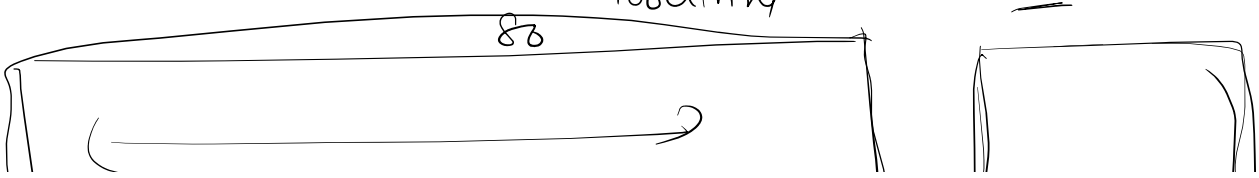
100 Samples

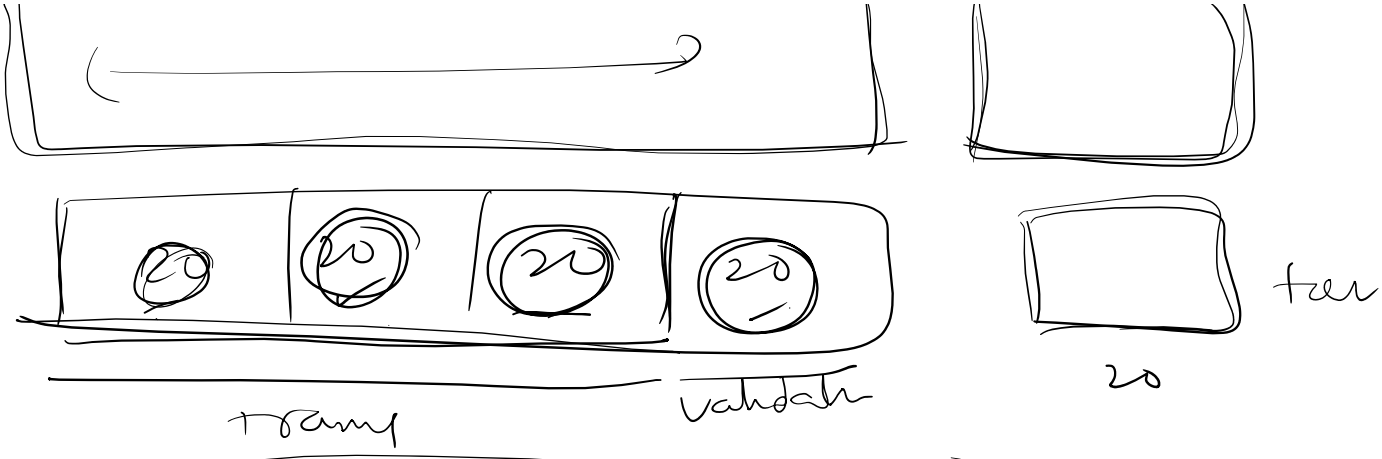
training

80

20 samples

testing





Model 1 — 80%

— 82%

— 80

— 78

$= \frac{80 + 82 + 80 + 78}{4}$

$= 80\%$

CV

20

Recap → definitions

→

→

→